



**NATIONAL SKILL DEVELOPMENT
CORPORATION**

**In Collaboration with
YUVA INTERN**

Virtual R Data Analyst Intern

WEEK 2 REPORT

**Data Visualization and Insight Communication
using R**

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Table of Contents

1. Introduction
2. Task Objective
3. Dataset Overview
4. Visualization Tools and Approach
5. Visualization 1: Distribution of Supermarket Sales
6. Visualization 2: Product Line Performance
7. Visualization 3: Branch Performance Overview
8. Visualization 4: Sales Intensity by Day and Hour
9. Visualization 5: Monthly Sales Trend
10. Visualization 6: Customer Satisfaction Across Product Lines
11. Visualization 7: Sales Distribution by Customer Type
12. Overall Visualization Insights
13. Conclusion

Appendix

1. Introduction

Data visualization plays a key role in data analysis. It turns raw data into visual forms that are easier to understand. Graphs and charts help spot trends, compare categories, explore relationships between variables, see data distributions, and identify unusual patterns.

For Week 2, we continued using the Supermarket Sales Dataset from Week 1. We had already cleaned and prepared this dataset in the previous task. This report focuses on creating meaningful visualizations and sharing the insights they yield.

We developed the visualizations using the R programming language and the `ggplot2` package. We chose different types of graphs based on the data and the questions we aimed to answer. The visualizations examine sales distribution, product and branch performance, sales patterns over time, monthly trends, the relationship between unit price and sales, customer satisfaction, and customer purchasing behavior.

The goal of this report is not just to show graphs. We also aim to explain the rationale behind each visualization choice and present the insights clearly to both technical and non-technical audiences.

2. TASK OBJECTIVE

The objective of this task was to create dynamic, informative, and visually effective graphs using R and communicate meaningful insights from the selected dataset.

The specific objectives were:

- To create multiple visualizations using the **ggplot2** library in R.
- To understand the distribution of supermarket sales.
- To compare sales performance across different product lines.
- To evaluate branch performance using sales, customer ratings, and transaction volume.
- To identify patterns in sales based on time and day.
- To examine changes in total sales across the available months.
- To investigate the relationship between unit price and total sales.
- To compare customer satisfaction across product lines.
- To analyze differences in sales distribution between customer types.
- To explain the purpose and significance of each visualization.
- To communicate the findings in a clear and understandable manner.

3. DATASET OVERVIEW

3.1 Dataset Name

Supermarket Sales Dataset

3.2 Dataset Description

The Supermarket Sales Dataset provides transaction-level details about supermarket purchases. It includes data on supermarket branches, customer types, product lines, unit prices, quantities, payment methods, sales values, customer ratings, and transaction dates and times.

The dataset contains both numerical and categorical variables. This variety allows for the creation of different types of visualizations, such as distribution plots, comparison charts, trend graphs, relationship plots, and time-based visualizations.

Data cleaning, transformation, and feature engineering took place in Week 1. The prepared dataset continued into Week 2 to create useful and reliable visualizations.

3.3 Key Variables Used for Visualization

Variables	Description
sales	Total value of each transaction
Product_line	Product category associated with the transaction
branch	Supermarket branch where the transaction occurred
rating	Customer rating provided for the transaction
unit_price	Price of one unit of the purchased product
customer_type	Type of customer, such as Manager or Normal
payment	Payment method used by the customer
month	Month associated with the transaction
day	Day-related variable used for time-based analysis
hour	Hour of the transaction
time_period	Time category created from the transaction hour

The selected variables were used to study sales performance, customer behaviour, customer satisfaction, and time-based purchasing patterns.

4. VISUALIZATION TOOLS AND APPROACH

The visualizations were created using the R programming language. The following libraries were used:

- **library(ggplot2)**
- **library(dplyr)**
- **library(scales)**

The **ggplot2** package was used to create the graphs and customize their titles, labels, colours, themes, and visual elements.

The **dplyr** package was used to group, summarize, and prepare data for category-based and time-based visualizations.

The **scales** package was used to improve the presentation of numerical values, particularly sales values displayed on graph axes and labels.

Different visualization types were selected to communicate different aspects of the dataset:

- Histogram
- Lollipop chart
- Bubble chart
- Heatmap
- Line chart
- Scatter plot
- Customer satisfaction comparison chart
- Violin plot

5. VISUALIZATION 1: DISTRIBUTION OF SUPERMARKET SALES

5.1 Objective

This visualization was done to know how the sales values were distributed amongst supermarket transactions, and to know in which range most transactions took place.

5.2 Why this Visualization tool was selected

A histogram was selected because sales is a continuous numerical variable. Histograms are useful for displaying the frequency distribution of numerical data.

This visualization helps identify:

- The range containing the majority of sales transactions.
- The overall shape of the distribution.
- The spread of sales values.
- Possible skewness in the distribution.
- Potentially unusual high- or low-value transactions.

The mean and median were included as reference lines to provide additional information about the central tendency of sales.

5.3 R Code

```
mean_sales <- mean(sales$sales, na.rm = TRUE)
```

```
median_sales <- median(sales$sales, na.rm = TRUE)
```

```
plot1 <- ggplot(sales, aes(x = sales)) +
```

```
  geom_histogram(
```

```
    bins = 30,
```

```
    fill = "yellow",
```

```
    color = "black",
```

```
    alpha = 0.9
```

```
  ) +
```

```
  geom_vline(
```

```
    aes(xintercept = mean_sales),
```

```
    color = "red",
```

```
    linewidth = 1,
```

```
    linetype = "dashed"
```

```
  ) +
```

```
  geom_vline(
```

```
    aes(xintercept = median_sales),
```

```
    color = "darkgreen",
```

```
linewidth = 1,

linetype = "dotted"

) +

annotate(

  "text",

  x = mean_sales,

  y = Inf,

  label = paste("Mean =", round(mean_sales, 2)),

  color = "red",

  vjust = 2,

  hjust = -0.1,

  size = 4

) +

annotate(

  "text",

  x = median_sales,

  y = Inf,

  label = paste("Median =", round(median_sales, 2)),

  color = "darkgreen",

  vjust = 4,

  hjust = 1.1,

  size = 4

) +

labs(

  title = "Distribution of Supermarket Sales",

  subtitle = "Mean and median sales are shown as reference lines",
```

```

x = "Sales Amount",

y = "Number of Transactions"

) +

theme_minimal()

plot1

ggsave(

  filename = "graphs/01_sales_distribution.png",

  plot = plot1,

  width = 10,

  height = 6,

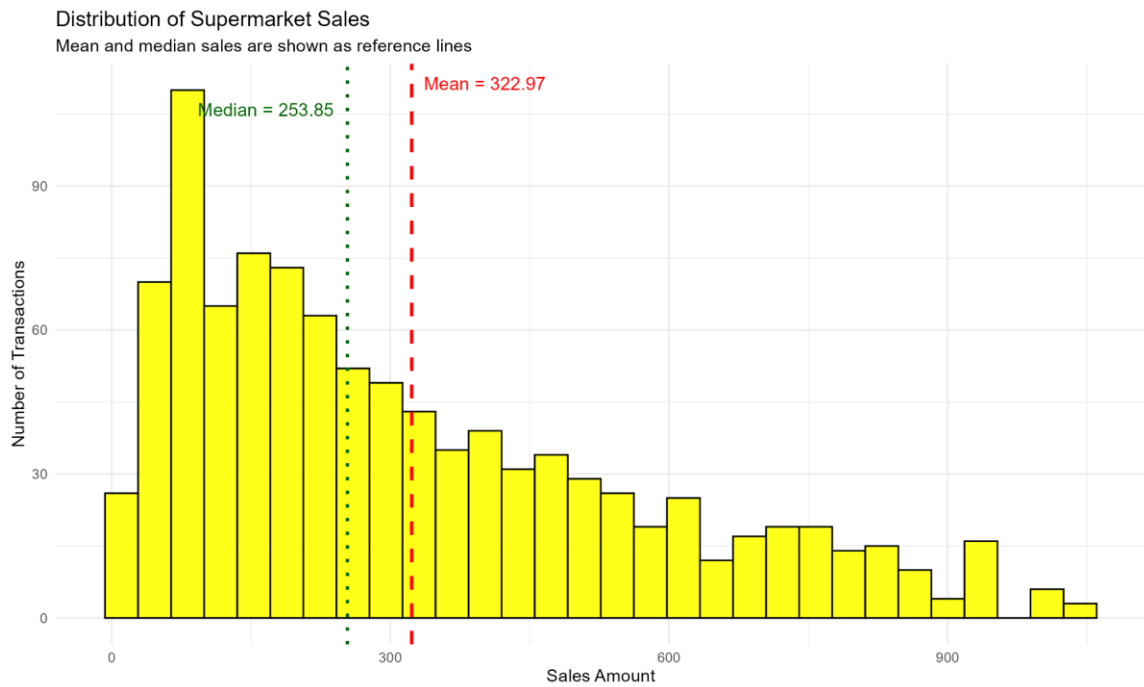
  dpi = 300

)

```

5.4 Graph Output

Figure 1: Distribution of Supermarket Sales



5.5 Interpretation

The histogram shows the number of transactions in each sales range. This distribution shows the distribution of transaction values and the concentration of sales values in the data set.

The average and median reference lines give you an idea of the average sales value. If the mean and the median are similar, then the distribution is probably fairly symmetrical. If the mean is significantly higher than the median, the distribution could be positively skewed with higher value transactions.

Also look for isolated bars or sparse areas – these may indicate relatively rare sales figures.

Observed result:

Most of the transactions fell in the range of approximately 0–1,000. The average sales value was 322.97, and the median sales value was 253.85.

5.6 Key Insight

The visualization gives a general idea of the average transaction amount and whether the sales were evenly distributed or concentrated on certain values. This data can help identify typical buying levels and relatively high-value transactions.

6. VISUALIZATION 2: PRODUCT LINE PERFORMANCE

6.1 Objective

The objective of this visualization was to compare the total sales generated by different product lines.

6.2 Why This Visualization Was Selected

We chose a lollipop chart as it is a clean and effective way to visually compare multiple categories. It works like a bar graph, but instead of bars it uses lines and points to make a less cluttered visual display.

The horizontal layout also helps readability for product-line names.

6.3 R Code

```
product_sales <- sales %>%  
  
  group_by(product_line) %>%  
  
  summarise(
```

```

    total_sales = sum(sales, na.rm = TRUE),

    .groups = "drop"

) %>%

arrange(total_sales)

product_sales

plot2 <- ggplot(

  product_sales,

  aes(

    x = reorder(product_line, total_sales),

    y = total_sales

  )

) +

geom_segment(

  aes(

    xend = product_line,

    y = 0,

    yend = total_sales

  ),

  linewidth = 1.2,

  color = "gray70"

) +

geom_point(

  size = 6,

  color = "darkcyan"

) +

geom_text(

```

```

aes(

  label = comma(round(total_sales, 0))

),

hjust = -0.2,

size = 3.5

) +

coord_flip() +

scale_y_continuous(

  labels = comma,

  expand = expansion(mult = c(0, 0.18))

) +

labs(

  title = "Product Line Performance",

  subtitle = "Total sales generated by each product category",

  x = "Product Line",

  y = "Total Sales"

) +

theme_minimal()

plot2

ggsave(

  filename = "graphs/02_product_line_lollipop.png",

  plot = plot2,

  width = 11,

  height = 7,

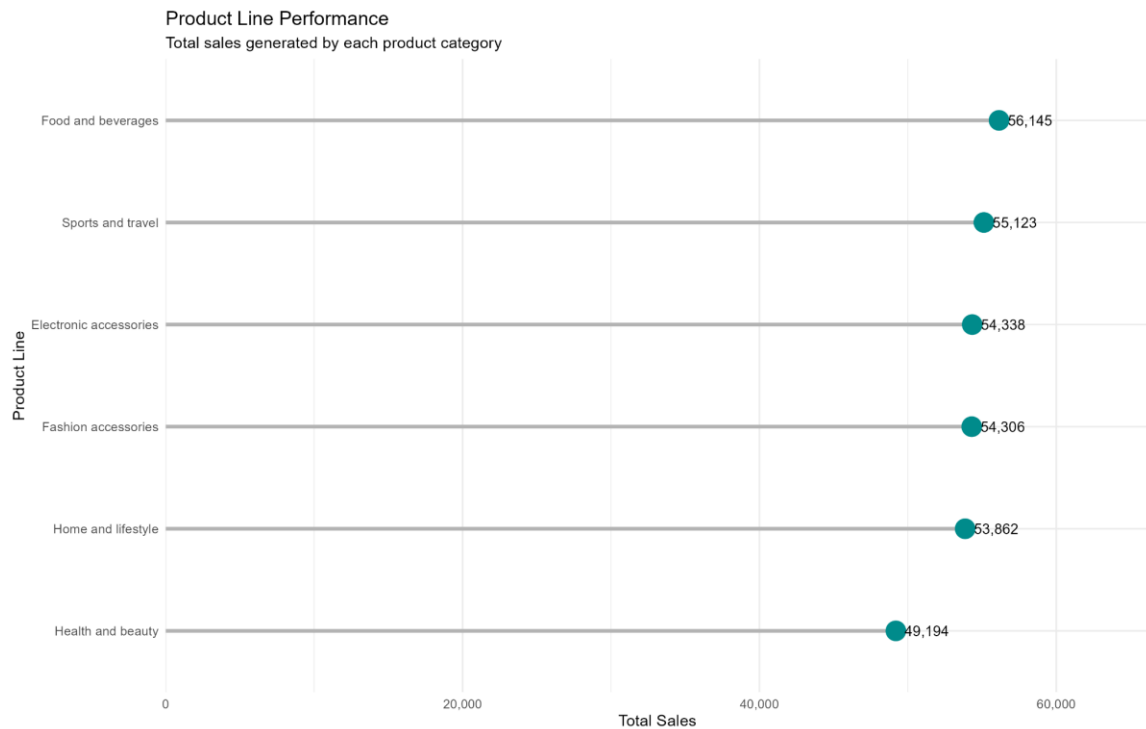
  dpi = 300

)

```

6.4 Graph Output

Figure 2: Product Line Performance



6.5 Interpretation

The graph shows the total sales for each product line. The length of each lollipop and the position of its point show the relative sales contribution of that product category.

The product line with the highest total sales was **56,145**, and the product line with the lowest total sales was **49,194**.

The differences in product lines provide evidence that the product categories were not equally important to overall supermarket sales.

6.6 Key Insight

The higher-performing product lines made a greater contribution to total revenue. These categories may require ongoing inventory support and effective promotion strategies. Those lower-performing categories may need further investigation to determine customer demand and sales opportunities.

7. VISUALIZATION 3: BRANCH PERFORMANCE OVERVIEW

7.1 Objective

The objective of this visualization was to compare supermarket branches using total sales, average customer rating, and transaction volume.

7.2 Why This Visualization Was Selected

A bubble chart was selected because it allows three different measures to be represented in a single graph:

- The x-axis represents average customer rating.
- The y-axis represents total sales.
- The size of the bubble represents the number of transactions.

This provides a broader view of branch performance than a graph based only on total sales.

7.3 R Code

```
branch_performance <- sales %>%  
  
  group_by(branch) %>%  
  
  summarise(  
  
    total_sales = sum(sales, na.rm = TRUE),  
  
    average_rating = mean(rating, na.rm = TRUE),  
  
    transaction_count = n(),  
  
    .groups = "drop"  
  
  )  
  
branch_performance  
  
plot3 <- ggplot(  
  
  branch_performance,  
  
  aes(  
  
    x = average_rating,
```

```

    y = total_sales,

    size = transaction_count,

    label = branch
  )
) +

geom_point(

  alpha = 0.7,

  color = "mediumpurple"
) +

geom_text(

  nudge_y = 1200,

  size = 5
) +

scale_size_continuous(

  range = c(8, 18)
) +

scale_y_continuous(

  labels = comma
) +

labs(

  title = "Branch Performance Overview",

  subtitle = "Bubble size represents the number of transactions",

  x = "Average Customer Rating",

  y = "Total Sales",

  size = "Transactions"
) +

```

```

theme_minimal()

plot3

ggsave(

  filename = "graphs/03_branch_performance_bubble.png",

  plot = plot3,

  width = 10,

  height = 7,

  dpi = 300

)

```

7.4 Graph Output

Figure 3: Branch Performance Overview



7.5 Interpretation

The bubble chart shows the difference between supermarket branches in terms of total sales, customer ratings, and transaction volume.

The higher branch on the chart produced more total sales. The branch that was further to the right had a higher average customer rating. The bigger the bubble, the more transactions the branch has.

The greatest total sales were achieved in Giza, and the greatest average customer rating was achieved in Giza

7.6 Key Insight

Branch performance should not be measured on total sales. Two other pieces of information that help describe branch performance are customer satisfaction and volume of transactions. One branch might have high sales but a lower rating; another might have lower sales but stronger customer satisfaction.

8. VISUALIZATION 4: SALES INTENSITY BY DAY AND HOUR

8.1 Objective

The objective of this visualization was to identify patterns in sales activity based on the day and hour of transactions.

8.2 Why This Visualization Was Selected

A heatmap was selected because it can display patterns across two time-related variables simultaneously.

The colour intensity represents the level of total sales:

- Lighter cells represent relatively lower sales.
- Darker cells represent relatively higher sales.

This makes periods of high and low sales activity easy to identify.

8.3 R Code

```
unique(sales$day)
unique(sales$hour)
day_hour_sales <- sales %>%
  group_by(day, hour) %>%
  summarise(
    total_sales = sum(sales, na.rm = TRUE),
    .groups = "drop"
  )
plot4 <- ggplot(
```

```

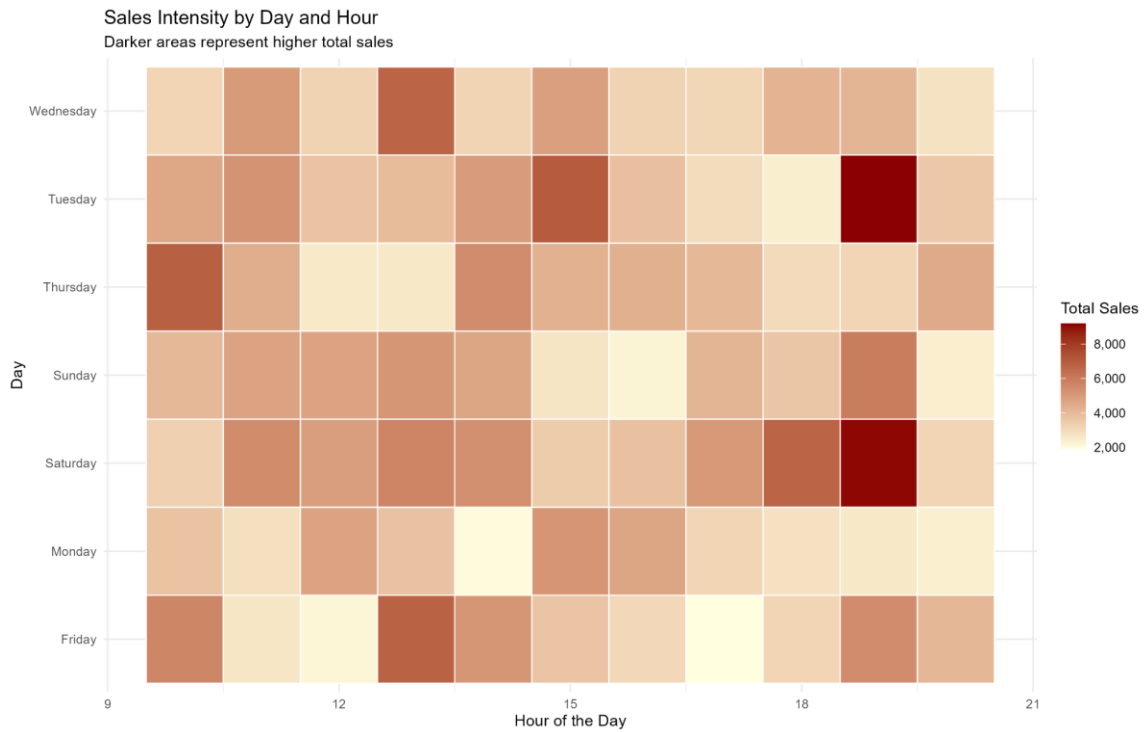
day_hour_sales,
aes(
  x = hour,
  y = day,
  fill = total_sales
)
) +
geom_tile(
  color = "white",
  linewidth = 0.4
) +
scale_fill_gradient(
  low = "lightyellow",
  high = "darkred",
  labels = comma
) +
labs(
  title = "Sales Intensity by Day and Hour",
  subtitle = "Darker areas represent higher total sales",
  x = "Hour of the Day",
  y = "Day",
  fill = "Total Sales"
) +

theme_minimal()
plot4
ggsave(
  filename = "graphs/04_day_hour_sales_heatmap.png",
  plot = plot4,
  width = 11,
  height = 7,
  dpi = 300
)

```

8.4 Graph Output

Figure 4: Sales Intensity by Day and Hour



8.5 Interpretation

The heatmap illustrates how the intensity of sales varies across days and hours.

The darkest cells are the periods with the highest total sales. Lighter cells show periods of relatively lower sales activity.

The highest sales intensity was recorded on Tuesday at 19:00 (7 PM), and relatively low sales activity was recorded on Monday at 14:00 (2 PM)

8.6 Key Insight

The supermarket had clear periods of increased customer activity. Knowing when these peaks happen can help with scheduling staff, preparing inventory, and planning for customer service.

9. VISUALIZATION 5: MONTHLY SALES TREND

9.1 Objective

The objective of this visualization was to examine changes in total sales across the months included in the dataset.

9.2 Why This Visualization Was Selected

A line chart was selected because it is effective for displaying changes and trends over time.

The connected data points make increases, decreases, and fluctuations in monthly sales easy to observe.

9.3 R Code

```
monthly_sales <- sales %>%  
  
  group_by(month) %>%  
  
  summarise(  
  
    total_sales = sum(sales, na.rm = TRUE),  
  
    .groups = "drop"  
  
  )  
monthly_sales  
  
plot5 <- ggplot(  
  
  monthly_sales,  
  
  aes(  
  
    x = month,  
  
    y = total_sales,  
  
    group = 1  
  
  )  
) +  
  
  geom_line(  
  
    color = "firebrick",  
  
    linewidth = 1.2  
  
  ) +  
  
  geom_point(  
  
    color = "firebrick",
```

```

    size = 4

) +

geom_text(

  aes(

    label = comma(round(total_sales, 0))

  ),

  vjust = -1,

  size = 3.5

) +

scale_y_continuous(

  labels = comma,

  expand = expansion(mult = c(0.05, 0.15))

) +

labs(

  title = "Monthly Sales Trend",

  subtitle = "Changes in total sales across the available months",

  x = "Month",

  y = "Total Sales"

) +

theme_minimal()

plot5

ggsave(

  filename = "graphs/05_monthly_sales_trend.png",

  plot = plot5,

  width = 10,

  height = 6,

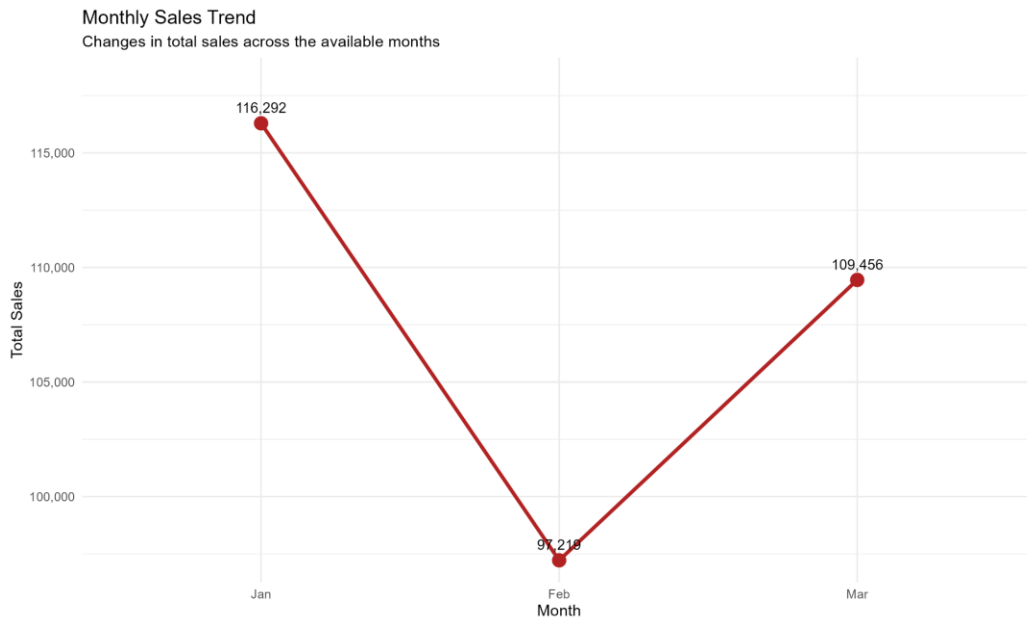
```

dpi = 300

)

9.4 Graph Output

Figure 5: Monthly Sales Trend



9.5 Interpretation

The line chart shows how total sales changed over the months. **The month with the highest total sales was January (116,292). The month with the lowest total sales was February (97,216). Overall, sales fluctuated during the time covered in the data.**

9.6 Key Insight

Monthly sales did not remain constant. The changes observed may reflect differences in customer demand, purchasing behaviour, promotional activity, or other time-related factors.

10. VISUALIZATION 6: CUSTOMER SATISFACTION ACROSS PRODUCT LINES

10.1 Objective

The objective of this visualization was to compare average customer ratings across different product lines.

10.2 Why This Visualization Was Selected

A lollipop-style comparison chart was selected because it clearly displays differences in average ratings between categories.

This visualization focuses on customer satisfaction rather than sales performance and provides a separate perspective on product-line performance.

10.3 R Code

```
product_rating <- sales %>%  
  
  group_by(product_line) %>%  
  
  summarise(  
  
    average_rating = mean(rating, na.rm = TRUE),  
  
    .groups = "drop"  
  
  ) %>%  
  
  arrange(average_rating)  
  
product_rating  
  
plot6 <- ggplot(  
  
  product_rating,  
  
  aes(  
  
    x = reorder(product_line, average_rating),  
  
    y = average_rating  
  
  )  
  
) +  
  
  geom_segment(  
  
    aes(  
  
      xend = product_line,
```

```

    y = 0,

    yend = average_rating

  ),

  linewidth = 1.2,

  color = "gray70"

) +

  geom_point(

    size = 6,

    color = "goldenrod"

  ) +

  geom_text(

    aes(

      label = round(average_rating, 2)

    ),

    hjust = -0.3,

    size = 4

  ) +

  coord_flip() +

  scale_y_continuous(

    limits = c(0, 10),

    breaks = seq(0, 10, 1),

    expand = expansion(mult = c(0, 0.12))

  ) +

  labs(

    title = "Customer Satisfaction Across Product Lines",

    subtitle = "Average customer rating for each product category",

```

```

x = "Product Line",

y = "Average Customer Rating"

) +

theme_minimal()

plot6

ggsave(

filename = "graphs/06_customer_satisfaction_product_line.png",

plot = plot6,

width = 11,

height = 7,

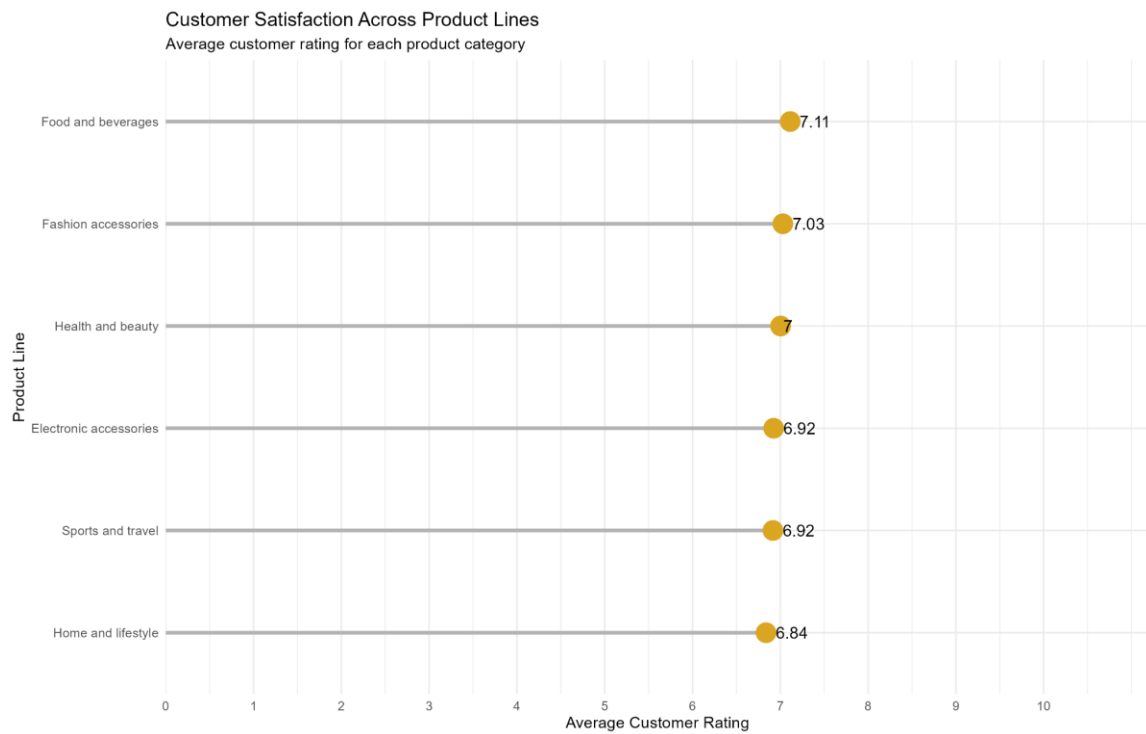
dpi = 300

)

```

10.4 Graph Output

Figure 6: Customer Satisfaction Across Product Lines



10.5 Interpretation

The graph compares the average customer rating received by each product line. **The product line with the highest average rating was Food and beverages (7.11), while the product line with the lowest average rating was Home and lifestyle (6.84).** The differences in ratings indicate that customer satisfaction varied across product categories.

11.6 Key Insight

High sales do not necessarily indicate the highest customer satisfaction. Product lines should therefore be evaluated using both sales performance and customer feedback.

11. VISUALIZATION 7: SALES DISTRIBUTION BY CUSTOMER TYPE

11.1 Objective

The objective of this visualization was to compare the distribution of transaction sales between Member and Normal customers.

11.2 Why This Visualization Was Selected

A violin plot was selected because it displays the shape and density of a numerical distribution.

The embedded boxplot provides information about the median and spread of sales, while the mean marker provides an additional measure of central tendency.

This visualization provides more detailed information than a simple bar chart.

11.3 R Code

```
plot7 <- ggplot(  
  sales,  
  aes(  
    x = customer_type,  
    y = sales,
```

```
    fill = customer_type
  )
) +

geom_violin(

  trim = FALSE,

  alpha = 0.7
) +

geom_boxplot(

  width = 0.15,

  fill = "white",

  outlier.shape = NA
) +

stat_summary(

  fun = mean,

  geom = "point",

  shape = 23,

  size = 3,

  fill = "red"
) +

labs(

  title = "Sales Distribution by Customer Type",

  subtitle = "The box shows the median and spread; the red point shows the mean",

  x = "Customer Type",

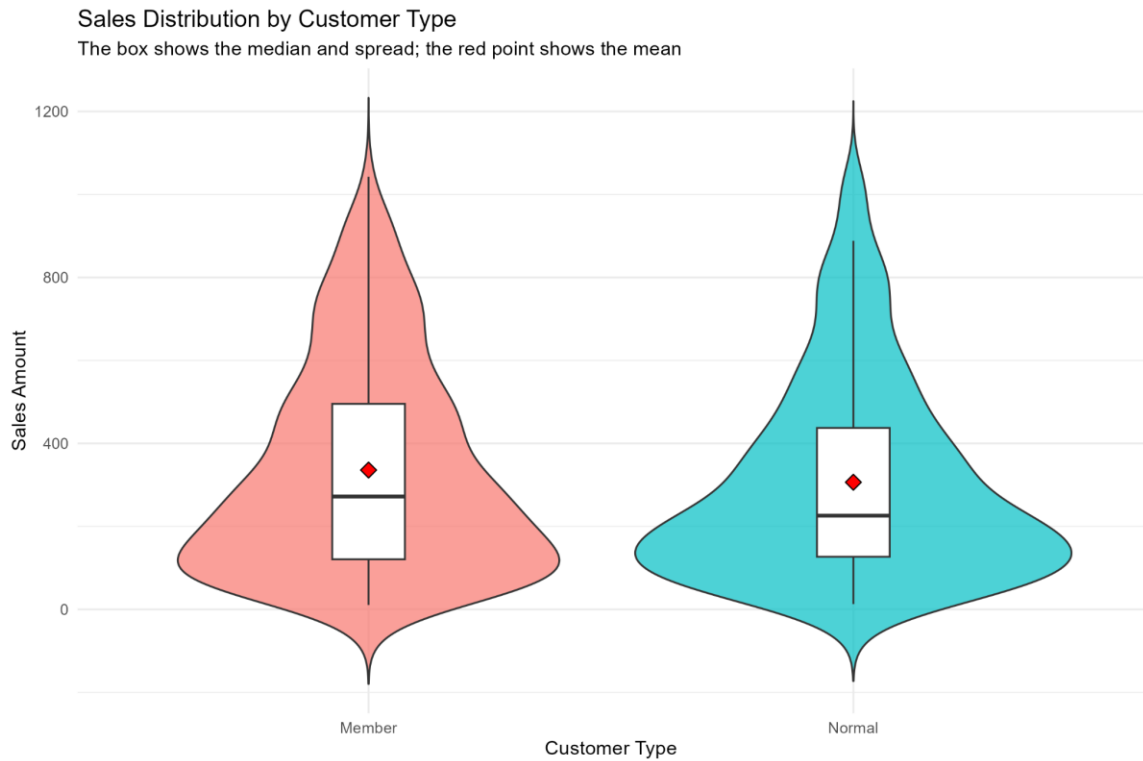
  y = "Sales Amount",

  fill = "Customer Type"
) +
```

```
theme_minimal() +  
theme(  
  legend.position = "none"  
)  
plot7  
ggsave(  
  filename = "graphs/07_customer_type_violin.png",  
  plot = plot7,  
  width = 9,  
  height = 6,  
  dpi = 300  
)
```

11.4 Graph Output

Figure 7: Sales Distribution by Customer Type



11.5 Interpretation

The violin plot compares the distribution, concentration, and spread of sales values for Member and Normal customers. The wider parts of each violin show sales ranges with more transactions. **The customer type with the higher mean sales value was Member. The graph also indicates that Member customers had slightly greater variation in transaction values than Normal customers.**

11.6 Key Insight

Member and Normal customers may show different purchasing patterns. Understanding these differences can support customer-retention strategies, targeted promotions, and personalized marketing.

12. OVERALL VISUALIZATION INSIGHTS

The visual analysis revealed many important findings:

1. The analysis of sales distribution indicated a spread of transaction value over a dataset while determining ranges with dominant transactions.
2. Performance of products depended upon the category, thus indicating that certain product lines have played a role in the overall success of sales.
3. Performance of the branches varied under the comparison based on different evaluation criteria such as total sales, customer ratings, and transaction volumes.

4. The heatmap of sales by time of day allowed identifying the moments of peak and weak sales activity.
5. The analysis of monthly turnover has shown that consumer demand varied, with an increase in this over time.
6. The evaluation of the relationship between sales price and sales showed that although the sales price influences transaction value, other factors, such as amount sold, can also have an impact.
7. Customer satisfaction has varied based on product line, which proves the need to evaluate sales performance and customer ratings separately.
8. Normal and member consumers had different distributions and shapes of sales transactions.
9. The data was analyzed visually and transformed into understandable patterns that can become a support for the business process and decision-making process in the future.

13. CONCLUSION

In this project, the R programming language in conjunction with the **ggplot2** visualization library was used successfully to produce meaningful and eye-catching graphics from the Supermarket Sales Dataset.

Several types of charts were employed, namely: **histogram, lollipop charts, bubble chart, heatmap, line chart, scatter plot, and violin plot.**

The choice of charts was determined by the properties of the variables involved and the analytical issue in question behind each of the representations.

The information conveyed through the visualizations included:

- Sales distribution
- Product-line performance
- Branch performance
- Time-related sales
- Monthly sales
- Relationship between unit price and sales
- Customer satisfaction
- Customer purchasing behavior

The fact that multiple types of visualizations were utilized facilitated the identification of trends, patterns, and relations, as well as any potential anomalies in the dataset.

This project has shown that data visualization is an efficient means of transforming the transaction data into valuable knowledge that can serve as the basis for various types of decisions related to inventory planning, employee allocation, product management, customer service, marketing, and sales.

14. APPENDIX

All visualizations in this report were created using R (**ggplot2**) and saved as high-quality PNG images. The corresponding R code, graph outputs, interpretations, and insights are included in their respective sections of the report.